EXP_2026Q2_CART_NUDGES · SRM CHECK · ENGINEER

GROWTH · CART · 2026-06-05 · X² P = 0.0340

Sample ratio is borderline.

The observed split drifted enough to warrant a look at the assignment service. Metrics can still be read, but the verdict carries an investigation note.

> WHAT THIS GATE DOES
>
> **Investigate assignment before relying on the verdict.**
>
> *A sample-ratio mismatch invalidates everything downstream — observed lift could just be a different population in each arm. The gate fires before any metric is read (R2).*

OBSERVED SPLIT

1303 / 1197

expected 1250 / 1250

X² STATISTIC	P-VALUE
4.49	0.0340
THRESHOLD	**VERDICT**
0.0005	WARNING

READING THIS CHECK

A small p-value here invalidates the rest of the analysis.

A p-value below the threshold but not catastrophically so suggests a small imbalance that may be sampling noise — or may reflect a real (small) assignment-service issue. Note it in the log; proceed cautiously.

COMPUTATION TRACE

srm_check is the only path to this number.

All SRM evaluations run through `agentxp.stats.srm.srm_check`; the function returns the trace below verbatim. Any change to the threshold or formula must come from a code change, not an ad-hoc call site (R4).

TEST	srm_chi_squared
OBSERVED	[1303, 1197]
EXPECTED	[1250, 1250]
CHI2_STAT	4.4944
P_VALUE	0.034006
THRESHOLD	0.0005

DF	1
VERDICT	WARNING
FORMULA	chi^2 = sum((obs − exp)^2 / exp) · p from chi2(df=k−1)
